A Solar-Powered LoRaWAN Framework for Real-Time Public Sanitation Monitoring and Demand-Driven Maintenance in Smart Cities

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***Abstract*—Cleanliness in urban sanitation facilities is a key problem of the city management. The old cleaning regimes are usually ineffective, and this results in the squandering of resources or the abandoned facilities. The paper suggests a self-sustaining and end to end Internet of Things (IoT) system to monitor the state of public toilets continuously. The system involves the use of a solar powered end-node that has ESP32 microcontroller, MQ-135 air quality sensors, and DHT11 environmental sensors. The transmission of data is through the LoRaWAN protocol, which gives the network a long range as well as a high penetration of the signal in congested urban areas. The node incorporates a solar energy harvesting unit to guarantee operational autonomy. The backend architecture uses Firebase Realtime Database to store the data and Firebase Cloud Messaging (FCM) to provide maintenance staff with instant push notifications in case of sanitation thresholds violation. The experiments illustrate that the system is reliable and can offer a demand-driven sanitation model that is scalable to fit the infrastructure of a smart city.**

***Keywords*- Air Quality Sensing, ESP32, Firebase Cloud Messaging (FCM), Internet of Things (IoT), LoRaWAN, Public Sanitation, Smart Cities, Solar Energy Harvesting.**

I.Introduction

City hygiene is an essential aspect of city health and decency. But in fast growing smart cities, it is becoming more unrealistic to ensure these facilities by manual inspection on a fixed-interval basis. These models do not take into consideration the changing patterns of usage and therefore, the hygiene begins to deteriorate during high traffic times. The concentration of both Ammonia (NH_3) and volatile organic compounds (VOC) is a major indicator of unsanitary conditions and possible health hazards.

Two major challenges to the implementation of remote monitoring systems in the public spaces include connection and power. Conventional wireless standards such as Wi-Fi are inappropriate since they have limited ranges and consume a lot of power. Moreover, it will be cost-prohibitive to ensure a stable wired power supply to all public toilets. This study will solve these issues by suggesting a Solar-Powered LoRaWAN-enabled architecture.

LoRaWAN (Long Range Wide Area Network) offers the penetration required at sub-GHz, which is required to pass through the concrete and ceramic mass of buildings in the public bathroom. Combining this with a specific solar energy harvesting circuit, the proposed IoT node will be fully energy independent, as it will never have to rely on the power grid. Combining it with Google Firebase, the system forms a unified pipeline between edge sensing and real-time alerting, allowing the municipal authorities to shift to a proactive, rather than reactive, approach to maintenance in the form of a clean-on-demand maintenance strategy.

The Smart City infrastructure is progressively becoming reliant on solid Internet of Things (IoT) models to control the utilities in the city [13]. In this field, one area that has been a major underserved application is in public sanitation where automated monitoring systems are replacing manual supervision [4]. Although the initial versions of IoT made use of short-range or high-power cellular schemes, the modern trend in the literature is to use Low Power Wide Area Networks (LPWAN) because of their peculiar ratio of range and energy efficiency [2], [12]. In particular, the LoRaWAN 2 Specification [1] offers the architectural support of the secure long-range communication that can penetrate through the dense concrete and ceramic material of the typical public restroom [10]. LoRaWAN has better performance in urban areas compared to rival technologies as has been shown in recent surveys [3] because it provides consistent connectivity at low signal-to-noise ratios. Nevertheless, the real scalability of such Edge Computing nodes, which can be based upon ESP32 SoC [6], is dependent on their energy independence. These nodes are made into self-sustaining assets of the urban environment by the incorporation of Solar Energy Harvesting units [5], which transform these devices that are battery limited into long-term environmental sensors. Moreover, only in case the data is actionable, the efficacy of air quality monitoring with the help of sensors, such as MQ-135 [8], will be achieved. Through the application of a serverless design based on Firebase Cloud Messaging (FCM) [7], sensor data are converted into real-time notifications. This change allows the so-called Demand-Driven maintenance models suggested in recent research [9], which guarantee that the maintenance of the standards of public health is carried out by proactive, data-driven sanitation policy.

1.1 Motivation

The main reason why this study was carried out is the urgent necessity to update the sanitation facilities of the

population in the framework of innovating smart cities. Traditional cleaning regimes are essentially data-blind, which can lead to the ineffective distribution of labor or, more importantly, the long-term presence of a dirty environment in busy bathrooms between cleaning sessions. Vital volatiles and the high levels of ammonia are not only a nuisance, but also first signs of bacterial growth and possible health hazards to the population. Moreover, the logistical issues related to delivering stable wired power and high-bandwidth to the distributed facilities of the population have historically constrained the implementation of real-time monitoring systems. Combining solar energy collection with the long-range/high-penetration LoRaWAN protocol, the proposed project will target the development of an entirely autonomous, install-and-forget sensing solution. The final goal is to move the model of municipal maintenance as a reactive and manual system to the proactive and demand-oriented system where cloud-based notification will ensure that any hygiene violations are addressed immediately and thus improve the efficiency of urban operations.

1.2 Proposed Work

The given work presents a full-fledged, solar-powered IoT system that is to be applied to the real-time monitoring and automated maintenance of the public sanitation facilities. A custom sensing node with a microcontroller, high-sensitivity gas and environmental sensors, is placed at the edge to measure the air quality and hygiene level. To eliminate the connectivity and power constraints of the urban infrastructure, the system employs LoRaWAN protocol to transmit low-power data over longer distances as well as to monitor a wide area and implements a monocrystalline solar harvesting unit to attain full energy independence. The whole data pipeline is handled by a LoRaWAN gateway that connects the physical sensors with a Firebase Realtime Database cloud-based application. One of the most important additions made by this work is the introduction of a threshold-based alerting engine using Firebase Cloud Messaging that converts raw environmental data to real-time push notifications to the maintenance staff. The given framework will help shift the traditional, time-based cleaning to a more data-driven, one-on-demand approach, maximizing allocation of municipal resources, as well as providing a decent level of consistent public health standards.

II.Methodology

This research methodology is based on the systematic approach to engineering, starting with a multi-layered system architecture, moving on to the physical prototype implementation and to a strict validation phase that will provide the reliability of the industrial grade.

A. System Design and Technical Architecture.

The proposed framework is founded on the four-layer Architecture of the IoT i.e., the Perception, Network, Middleware and Application layers. The perception layer has the ESP32 microcontroller in the center since it is dual-core with highly granular power-management capabilities, both of which are necessary in long-term remote deployments. This unit is linked with a high-sensitivity MQ-135 gas detector which relies on a layer of tin-dioxide (SnO_2) semiconductor to detect ammonia (NH_3), sulfides and other volatile organic compounds (VOCs) to indicate poor sanitation. The DHT11 sensor to complement this would give information on the ambient atmosphere to apply in correlating the intensity of odor to either the high-humidity or the high-temperature environment.

The system will be based on the LoRaWAN Class A protocol to support the communication backbone, and SX1278 LoRa transceiver. This sub-GHz technology was chosen with a purpose to fight the so-called Faraday cage Effect that is usually created by the concrete and ceramic tiles of the restrooms of people that usually blocks 2.4 GHz Wi-Fi signals. In order to achieve energy-autonomous behavior of the system, the design includes a 6V/250mA monocrystalline solar panel and a CN3065 Lithium-ion charge management IC. This design will sustain a Li-ion battery of 18650, with a capacity of 2500mAh, which will be able to supply power continuously during consecutive days of low solar irradiance.

B. Prototype Construction and Software Implementation.

The prototype was constructed with proper implementation of hardware components on a single control board to minimize the phenomenon of electromagnetic interference (EMI) and signal attenuation. The 48 hours long burn-in or pre-heating of the MQ-135 sensor was one of the most important steps in the assembly process as it is necessary to stabilize the sensing element, as well as ensure the linearity of the readings of gas concentrations. The ESP32 firmware was developed based on the Arduino platform together with LMIC (LoRaWAN-MAC-in-C) library in order to support more sophisticated network tasks like Over-the-Air Activation (OTAA) and adaptive data rate (ADR) adjustments.

The gadget was enclosed in an IP65 weatherproof cover with the solar panel tilted at 15 degrees to enable the gadget to get as much photons as possible. The backend server was built on the serverless architecture on the cloud side. The network server was used to decode the received hexadecimal payloads with the use of The Things Network (TTN). These decoded values were then forwarded using a RESTful Webhook to a Firebase Realtime Database. The Cloud Function was applied to search the database and identify violation of the threshold to enable real-time alerting through the provision of real-time notification of the violation. The role acts as the system decision-maker, and it makes calls to the Firebase Cloud Messaging (FCM) API to transmit high-priority push notifications to the web dashboard of the administrator.

C. Testing, Verification and performance Analysis.

To test and validate the system was done via a three-stage experimental protocol to ascertain readiness of the system to be deployed into a smart city. To begin with, Signal Propagation Testing was performed through inserting the node into a structural public toilet and recording the Received

Signal Strength Indicator (RSSI) and Signal-to-Noise ratio (SNR) at different distances. The measurements confirmed the steady data packets that are received by the gateway to 2.5 km in a dense urban environment, which testifies to the increased penetration of the LoRa modulation.

Second, the Energy Autonomy was assessed with the help of a sustained 72 hours of power-cycling test. It was observed that the voltage over the battery was measured with an analog-to-digital converter (ADC) of the ESP32 that the current in the solar charger could sustain a fixed replenishment of the energy used during the active stages of transmissions and the 10 micro-amps of deep-sleep stages. Lastly, Operational Latency was also assessed by modeling a hygiene violation with an ammonia source that was concentrated. The system proved to be very responsive in terms of notification pipeline and the average latency, between the time the gas threshold was surpassed and the time the push notification was received by the web dashboard was 1.45 seconds on average. This is a real-time response which is an indication that the system is capable of supporting real-time, demand-based maintenance strategies.

III. System Architecture Proposal

The suggested system architecture will support the smooth data stream between the real-life sensing environment and the cloud-based alert system. It employs a 4-layered IoT architecture, the Perception Layer, the Network Layer, and the Application Layer which is powered by a powerful Power Management Unit.

A.Perception Layer and Edge Intelligence.

The Perception Layer is the eyes and ears of the system that is applied in the public sanitation facility. It is based on ESP32 microcontroller, which is a highly powerful SoC, which can operate a wide range of sensors at ultra-low power consumption. It has a special MQ-135 semiconductor sensor that is incorporated in this layer and it uses a sensitive material of tin-dioxide (SnO 2) to detect the concentration of Ammonia (NH3) and other nitrogenous substances that may be a result of poor hygiene. At the same time, a DHT11 sensor will be used to determine the ambient temperature and humidity that will provide the environmental context that can affect the volatility of the gas. The ESP32 includes an ESP32-Edge Intelligence which transforms the raw analogue signals into a digital form and uses a calibration algorithm to convert the values of resistance into parts per million (PPM) and sends it.

B. Energy independent Power Management Unit.

To allow the system to be utilized in remote public places that are not part of the electrical system, an extra unit is introduced known as Power Management Unit (PMU). The source of energy used in this sub-system is a 6V Monocrystalline Solar Panel which is the most efficient in the gathering of diffused sunlight in cities. This energy is controlled by a CN3065 Solar Charge Controller using a constant-current/constant-voltage (CC/CV) charging profile to ensure a 2500mAh Lithium-ion battery is safely charged. This unit also makes sure that the end-node is operational 24/7 and the solar harvesting during the daylight hours is sufficient to compensate the power used during active LoRaWAN uplink periods and during the deep-sleep of the micro-ampere deep-sleep periods.

C. Topology of the LoRaWAN Network and Gateway.

The Network Layer closes the gap between the sanitation facility and the internet with the help of the LoRaWAN (Long Range Wide Area Network) protocol. The sensor information is encrypted into a small payload and an SX1278 transceiver on the end-node packages it into a payload. This message is transmitted by the sub-GHz radio waves (868/915 MHz) to a LoRaWAN Gateway. Such a topology is highly penetrating as opposed to traditional Wi-Fi in that it is capable of penetrating through several concrete walls and hitting the gateway which may be as far as 2.5 kilometers. The gateway acts as a transparent cut tunnel and forwards the packets of the received RF to The Things Network (TTN) via an Ethernet or cellular backhaul to be authenticated and removed on a network basis.

D. Cloud Data Engine and Notification Engine.

The last level in the architecture is the Application Layer which operates on Google FireBase. After receiving the LoRaWAN payload by the network server, the payload is sent via a RESTful Webhook to the Firebase Realtime Database (RTDB) where it is stored with a rigid timestamp to examine it over time. At the same time, an inflow of data is checked by a serverless Firebase Cloud Function. Once the measured level of the gas is higher than the established level of sanitation (e.g. 300 PPM) the function activates Firebase Cloud Messaging (FCM) API. This engine sends an instant high-priority push notification to the Web dashboard of the administrator to convert raw data into a viable maintenance request.

IV. Findings and Discussion

The LoRaWAN sanitation system is a solar-powered system that was tested over a 72 hour period of continuous operation in a busy public restroom. The outcomes can be categorized into network performance, energy sustainability and sensor to alert latency.

A. LoRaWAN Signal Reliability and Propagation.

The principal aim of the network testing was to test the ability of sub-GHz signal to penetrate the dense structural substance of a public toilet, reinforced concrete and ceramic tiles. The transmission of data packets was done at SF7 bandwidth of 125 kHz.

- **Range Performance:** The communication range of 2.5 km was constant at the gateway.
- **Signal Quality:** Signal to Noise Ratio (SNR) was at the average of greater than -5 dB and Received Signal Strength Indicator (RSSI) at the average of -108 dBm which has ensured that Packet Delivery Ratio (PDR) was at 96.4%. Such results confirm that LoRaWAN is far more reliable than the 2.4 GHz Wi-Fi when it comes to the application of the sanitation monitoring in the city in deep-indoor locations.

B. Solar Gathering Effectiveness and Energy Independence.

The battery voltage was monitored in the test cycles that used the maximum solar irradiance, and the nighttime use to verify the install-and-forget feature of the node.

- **Consumption Profile:** ESP32 active LoRa transmission (2 seconds) used about 120mA and went down to 10muA when in deep-sleep (10 minutes) mode.
- **Harvesting Rate:** The 6 V monocrystalline panel gave an average charge current of 180mA during the day.
- **Net Result:** The amount of energy collected on 6 sunny hours was enough to energize the node on 72 hours of uninterrupted use which essentially confirms the fact that the system can also operate on consecutive cloudy days without any recharging.

C. Sensor Calibration and Odor Correlation.

The data of the MQ-135 sensor was calculated to align the trend of facility use with the degree of gas concentration.

- **Baseline vs peak:** At the time of a low concentration of ammonia (3:00 AM - 5:00 AM) it was found that the amount of ammonia was approximately 85 PPM. The level rose in the linear concentration on the peak hours (8:00 AM -10:00 AM) and was frequently exceeding the 300 PPM limit.
- **Environmental Impact:** The measured result of the DHT11 showed a slight enhancement in the perception of odor intensity in the humid environment (>75%), which showed that the multi-sensor method was required to evaluate the hygiene correctly.

D. Latency of notification and responsiveness of the system.

The response to the project in terms of demand is the Demand-Driven which depends on the pace of the notification pipeline. The measurements of the "Edge-to-End" latency, which is the time taken between the point when the sensor reported a threshold violation and when a push notification was received on the web dashboard, were taken.

- **Average Latency:** 1.45 seconds.
- **Breakdown of the Components:** LoRaWAN uplink (1.1s), Firebase Cloud Function execution (0.2s), and Firebase Cloud Messaging (FCM) delivery (0.15s). This kind of low reaction time means that the municipal authorities will be able to dispatch cleaning to clean up a facility at the time it becomes unhygienic and not at a set and inefficient time.

Metric	Observed Value	Performance Status
Max Communication Range	2.5 km	Excellent (Urban)
Packet Delivery Ratio	96.4%	High Reliability
Sleep Current	10μA	Ultra-Low Power
Alert Latency	1.45s	Real-Time
Battery Stability	+0.2V (Daily Net)	Fully Autonomous

TABLE I: Summary of System Performance Metrics
The experimental evaluation of the integrated system confirms its suitability for large-scale urban deployment. As summarized in **Table I**, the system's performance was measured across five critical dimensions: range, reliability, power efficiency, latency, and energy autonomy.

V. Findings and Future Research.

The conclusion will be able to recap off the technical successes of the project and how this research can be further developed in the future.

A. Conclusion

The study has been able to illustrate how to design and deploy a self-sustained, IoT infrastructure based on LoRaWAN to monitor real-time sanitation in the city. The system combines a solar energy harvesting with an ESP32-based sensing node, making the system entirely energy independent, overcoming the power constraints of conventional urban infrastructure. The use of the LoRaWAN protocol was very effective in the dense structural settings, and it ensured a consistent connection coverage of a distance of 2.5 km and a packet delivery rate of 96.4 percent. Besides, the Firebase Cloud Messaging integration allowed a near-real-time notification pipeline and a latency of just 1.45 seconds. Such transition between a fixed-time cleaning system to a proactive, demand-oriented sanitation system not only maximizes municipal resources but also guarantees a high level of hygiene in the population of smart cities.

B. Future Scope

Although the existing framework offers a sound solution to the area of hygiene monitoring, there are a number of opportunities to improve the approach in the future:

- **TinyML Integration:** In future versions, machine learning models can be realized on the ESP32 (Edge AI) to predict cleaning needs, based on past data trends and frequency of use, instead of using a hard-coded threshold.
- **Extension of Sensor Store:** It would be nice to add more sensors like waste-level sensors to sensors in bins or water-flow sensors to monitor utility consumption and have a more comprehensive perspective of facility maintenance.
- **Mobile Actuation:** This could be further optimized by creating a dedicated mobile application to clean the staff that takes into account GPS navigation to the facility that activated the alert.
- **Hybrid Connectivity:** The topology of a multi-gateway mesh may be explored to achieve an even greater range, to monitor public facilities in very remote or geographically challenging cities.
- **Advanced energy analytics:** To further increase the efficiency of energy collection in low-light environments, more advanced Maximum Power Point Tracking (MPPT) algorithms can be implemented in the solar charging circuit.

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